Parardha Dhar

Email: parardhadhar@gmail.com | **Phone:** +91 8918858767 | **Portfolio:** parardha.vercel.app | **GitHub:** github.com/parardhadhar | **LinkedIn:** linkedin.com/in/parardhadhar

PROFESSIONAL SUMMARY

Computer Science and Engineering student at VIT Bhopal University (GPA: 8.45, graduating 2028) with hands-on experience building large-scale, resilient software systems across artificial intelligence, machine learning, full-stack web development, and cloud infrastructure. Proven ability to own projects end-to-end — from design and coding through testing and deployment — including a production platform that served 1,800+ users and sustained 1,000+ clicks per second. Experienced in Python, Java, CI/CD, Docker, Kubernetes, and agile methodologies. Passionate about using technology to solve real-world problems and create innovative solutions that make a difference.

EDUCATION

VIT Bhopal University *Sep 2024 – 2028 | Bhopal, India*
Bachelor of Technology in Computer Science and Engineering **| GPA: 8.45 / 10.0**

Don Bosco School, Darjeeling *— Graduated May 2024*

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL, CUDA
ML & Artificial Intelligence: PyTorch, TensorFlow, scikit-learn, OpenCV, Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), agentic workflows, conversational AI *(familiar with ecosystem architectures like NVIDIA NIM/NeMo)*
Web & Full-Stack Development: Next.js, React, Node.js, REST API Design, Microservices, Mobile-Responsive Design, Vercel
Cloud & DevOps: Oracle Cloud Infrastructure (OCI), Docker, Kubernetes, CI/CD, GitHub Actions, Application Resiliency, Security Best Practices
Databases: Supabase, Firebase, SQL, Relational Databases, Data Warehousing Concepts
Software Engineering: Data Structures and Algorithms, Object-Oriented Programming, System Design, Agile Methodologies, Software Testing, Business Analysis, Distributed Systems, Big Data Processing

EXPERIENCE

Machine Learning Research Intern—Computational Materials Engineering | **IISc Bangalore** | June 2026-Present

- Architected "TEM AI Studio," a desktop SaaS OS bridging crystallographic physics and Generative AI, reducing manual defect calculation time by 98% (from 2+ hours to under 15 seconds per micrograph).
- Engineered a custom AI Inference Core using PyTorch, fine-tuning Diffusion Models (DDPM) to reconstruct lattice images, which improved downstream YOLOv11 defect detection accuracy by 45% on highly noisy atomic structures.
- Developed a High-Performance Physics Engine using OpenCV, executing real-time 2D Fast Fourier Transforms (FFT) and geometric Burgers vector derivations with 99% mathematical precision across 50+ megapixel scans.
- Integrated an Agentic LLM Copilot (Llama-3.3-Nemotron) via Server-Sent Events (SSE), reducing user-click fatigue by 60% by allowing the AI to autonomously trigger UI actions and generate scientific reports.
- Deployed a secure, decoupled architecture handling 3 concurrent data pipelines (Physics, Diffusion, LLM) utilizing a strict RESTful FastAPI backend and an Electron frontend hardened with ASAR packaging and context isolation.

PROJECTS

TEM AI Studio: Agentic OS for Crystallography
React, Electron, FastAPI, PyTorch, Diffusion Models, YOLO

- Deployed a secure, decoupled architecture handling 3 concurrent data pipelines (Physics, Diffusion, LLM) utilizing a strict RESTful FastAPI backend and an Electron frontend hardened with ASAR packaging and context isolation.

- Implemented a denoising diffusion pipeline (DDPM) to synthetically reconstruct noisy micrographs, boosting the downstream YOLOv11 and SAM pixel segmentation accuracy by 45% on raw atomic scans.
- Engineered a real-time physics engine using OpenCV, processing 50+ megapixel datasets to instantly calculate 2D-FFT reciprocal space masks and geometric Burgers vectors with 99% mathematical precision.
- Embedded an autonomous AI Copilot (Llama-3.3-Nemotron) that streams live JSON payloads to trigger UI actions, reducing user-click fatigue by 60% and automating the generation of exportable scientific reports.

Agentic Document Insight Engine—Conversational *AI & RAG System | 2024*

Python, LangChain, Hugging Face, FAISS, Big Data Processing, Nvidia NIM

- Built an intelligent conversational AI agent utilizing LangChain and Hugging Face to automate multi-document analysis and workflows
- Designed a high-performance RAG framework, chunking and embedding unstructured data into a FAISS vector store to optimize low-latency assistant responses.
- Containerized and scaled inference endpoints for real-time interaction via REST APIs, utilizing architecture patterns directly transferable to optimized microservice deployments.
- Upgraded conversational AI backend to utilize NVIDIA NIM inference endpoints for accelerated LLM performance.

PhishGuard—Real-Time Cybersecurity Threat Detection System 2024

Python, scikit-learn, Flask, Docker, REST API | github.com/parardhadhar/phishguard

- Developed a machine learning URL classification system applying cybersecurity and security best practices to detect phishing threats in real time; containerized with Docker and exposed via REST API for integration into enterprise security pipelines
- Prioritized a low false-positive rate over recall based on business analysis—incorrect security flags damage user trust more than occasional misses, a deliberate product decision that reflects real-world software improvement thinking.

VRGC Community Hub — Full-Stack Web Platform · VR & Gaming Club, VIT Bhopal 2025

Next.js, React, Supabase, Firebase, Vercel, REST APIs, Cl/CD, Cloud Deployment | vrgc.vercel.app

- Designed and delivered a large-scale, resilient full-stack platform end-to-end—handling all design, coding, testing, and cloud deployment solo while managing live operations during a high-traffic production event
- Scaled the system to serve 1,800+ unique visitors and 8,300+ page views; the ticketing and interaction layer sustained 1,000+ clicks per second without bottlenecking, validating the application resiliency design under real load.
- Achieved a 96/100 Real Experience Score on Vercel Speed Insights under peak load (First Contentful Paint 1.32s, Largest Contentful Paint 1.48s) by optimizing asset delivery for 800+ concurrent live stream viewers.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 — Generative AI Professional (Oracle)
AI/ML for Geo Data Analysis (ISRO — Indian Space Research Organisation)
Supervised Machine Learning: Regression & Classification (Deep Learning). AI) **Machine Learning with Python** (IBM)
Introduction to Security Principles in Cloud Computing (Google Cloud)
Competitive Programming: 100+ problems solved on LeetCode, Codeforces, GeeksForGeeks, and HackerRank—strong in data structures and algorithms, dynamic programming, graphs, recursion, and greedy algorithms. Actively growing.

RELEVANT COURSEWORK

Data Structures and Algorithms · Operating Systems · Artificial Intelligence · Machine Learning · Computer Graphics · Database Management Systems · Unix Tools and Scripting · Real-Time Systems · Numerical Computing · Software Engineering